# Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

**Cell Type** Mammalian

**Species Used** Mouse, A-9, derivative of mouse L cell (contains human chromosomes)

**Molecules Electroporated** DNA, various linear constructs of human sequences & selectable markers.

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**Before the Pulse**

**Cell Growth Medium** DMEM (GIBCO/BRL, Sigma)

**Growth Phase at Harvest** Not given

**Pre-pulse Incubation** 10 min. ice

**Wash Solution** Phosphate Buffered Saline, without Ca++, Mg++

**The Pulse**

**Instruments Used** Not given

**Electroporation Temperature** Cuvette on ice just prior to pulse

**Electroporation Medium*** Phosphate Buffered Saline, without Ca++, Mg++

**Cell Density** 10 (7) cells / ml

**Volume of Cells** 0.8 ml

**DNA Concentration** 1 µg /1 µl

**DNA Resuspension Buffer** water

**Volume of DNA** 10 µl

**Cuvette Gap** 0.4 cm

**Voltage** 0.45 kV

**Field Strength** 1.125 kV/cm

**Capacitor** 500 µF and 960 µF

**Resistor** (Pulse Controller) Ω none

**Time Constant** Not given

**After the Pulse**

**Outgrowth Medium** DMEM

**Outgrowth Temperature** 37 °C

**Length of Incubation** 48 hours

**Selection Method or Assay Used** Hygromycin, G418, or MX

**Electroporation Efficiency** Low

**Per Cent Survival** Not given

**Relevant Publications and/or Comments**

**Note:** exponential values designated in parentheses.

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**Name of Submittor** Keith Rosenbach

**Institution Address** UMDNJ
Microbiology Department
185 South Orange Ave
Newark, NJ 07103


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